Chapter 29

An Introduction to Chromatographic Separations

***Fundamentals of Analytical Chemistry*, Skoog, et al., 10th Ed., 2022**

Chapter 29 Problems:

29-3, 29-5, 29-7, 29-8, 29-9, 29-10, 29-24, 29-26a

Russian botanist Mikhail Tsweet separated plant pigments by passing solutions through a glass tube packed with calcium carbonate. The species separated as colored bands on the column

Chroma comes from the Greek meaning color and graphein meaning writing

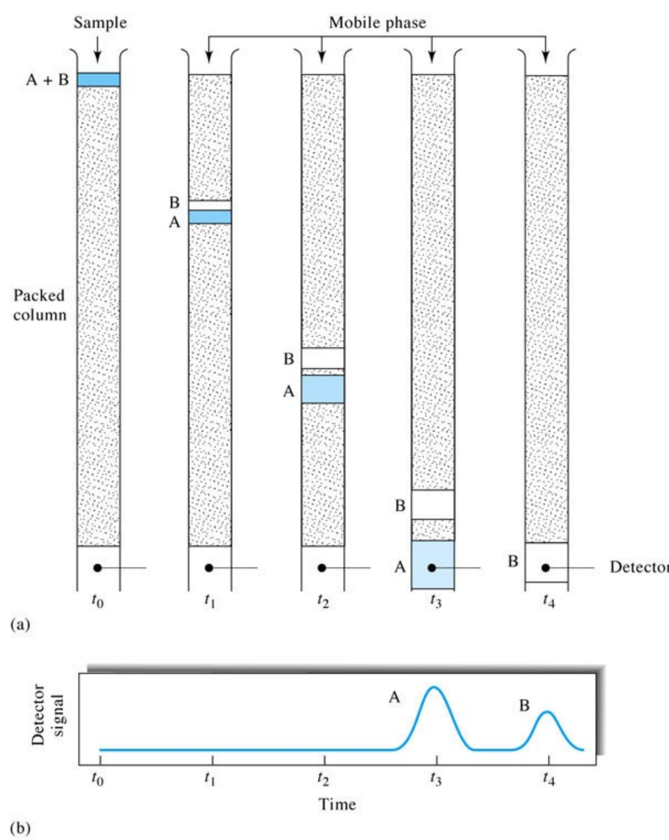
All chromatography methods involve the separations of analytes in a mobile phase through contact with a stationary phase material

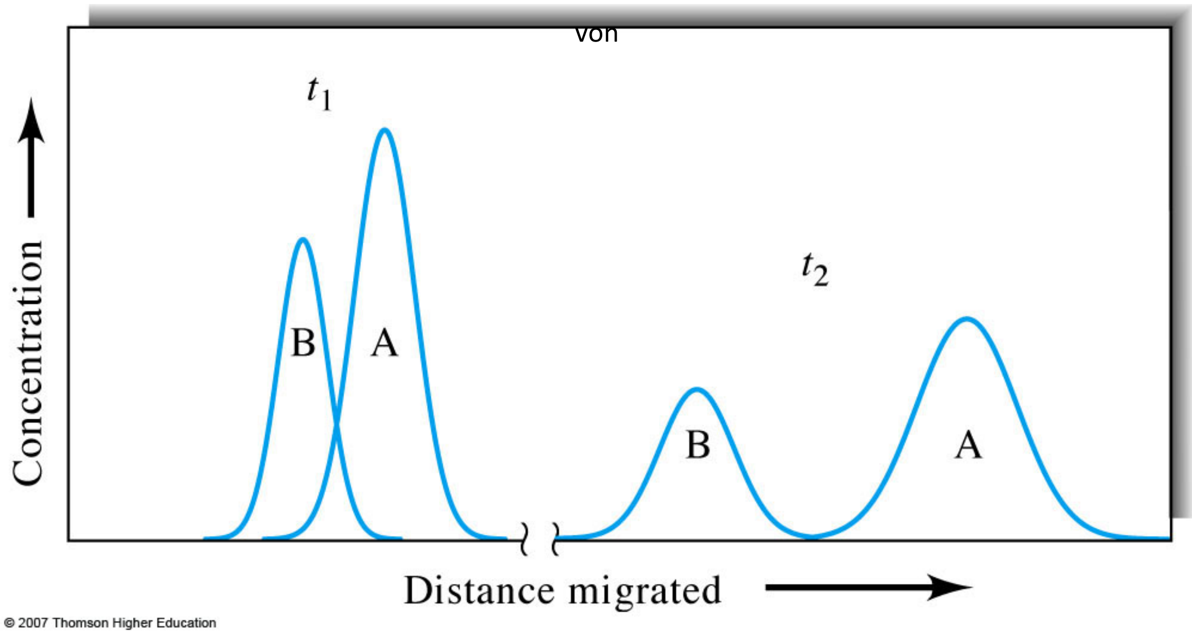
Each phase is immiscible in the other

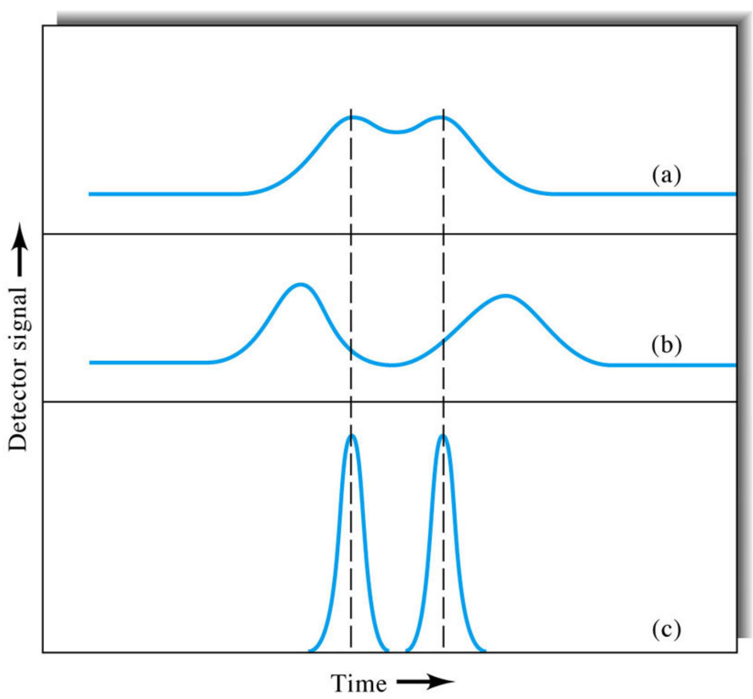
In column chromatography stationary phase is held in a narrow tube through which the mobile phase is forced under pressure

The analytes repeatedly interact with the stationary phase (partitioning) which gradually separates the analytes into bands in the mobile phase.

The analytes that interact the least with the column material come out first.







A) Original chromatograph with overlapping peaks
B) Improvement brought about by an increase in band separation
C) Improvement brought about by a decrease in bandwidth

The positions of a peak in the chromatograph can be used to identify an analyte

The area under the peak can be used to quantify an analyte

Peak area is inversely related to the flowrate of the mobile phase

Chromatography Classifications- based on the phase of the mobile and stationary components.

Mobile phase-can be a gas or a liquid

Stationary phase can only be a liquid or a solid

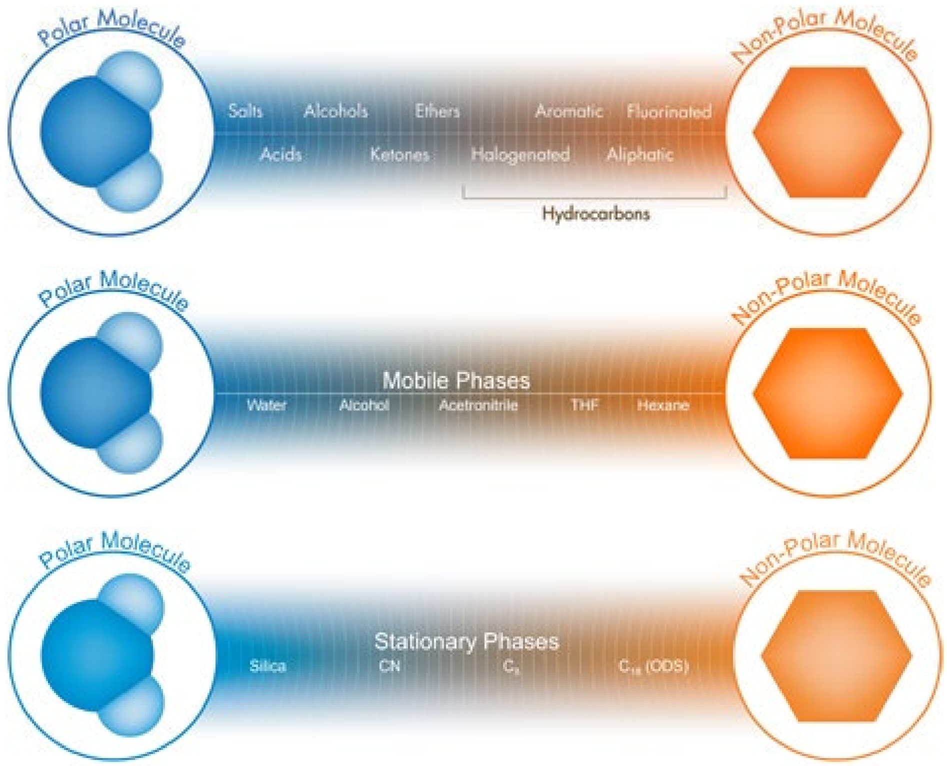
In gas-liquid chromatography, the mobile phase is a gas, whereas in liquid-liquid chromatography, it is a liquid.

TABLE 31-4

Classification of Column Chromatographic Methods

General Classification	Specific Method	Stationary Phase	Type of Equilibrium
1. Gas chromatography (GC)	a. Gas-liquid (GLC)	Liquid adsorbed or bonded to a solid surface	Partition between gas and liquid
	b. Gas-solid	Solid	Adsorption
2. Liquid Chromatography (LC)	a. Liquid-liquid, or partition	Liquid adsorbed or bonded to a solid surface	Partition between immiscible liquids
	b. Liquid-solid, or adsorption	Solid	Adsorption
	c. Ion exchange	Ion-exchange resin	Ion exchange
	d. Size exclusion	Liquid in interstices of a polymeric solid	Partition/sieving
	e. Affinity	Group specific liquid bonded to a solid surface	Partition between surface liquid and mobile liquid
3. Supercritical fluid chromatography (SFC) (mobile phase: supercritical fluid)		Organic species bonded to a solid surface	Partition between supercritical fluid and bonded surface

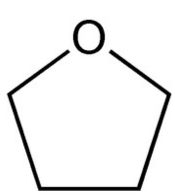
Size exclusion chromatography is useful of separating polymers based on MM.



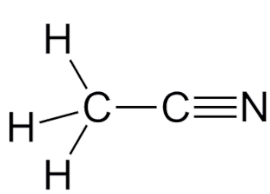
Separation Mode	Stationary Phase [particle]	Mobile Phase [solvent]
Normal phase	Polar	Non-polar
Reversed phase	Non-polar	Polar

Hydrogen Bonding (H-O, H-N, H-F)

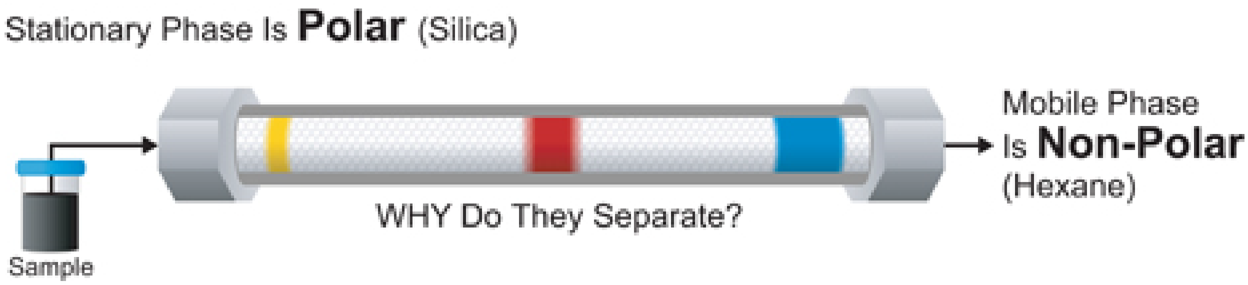
THF = tetrahydrofuran = C_3H_8O



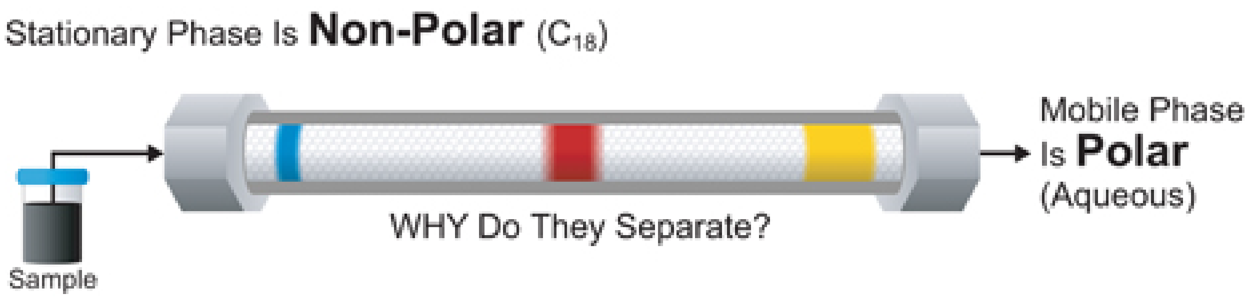
Acetonitrile = C_2H_3N



Normal Phase (*because Tsweet used chalk (carbonate as his stationary phase which is polar)*



Reverse Phase



Chromatographic Behavior

The way an analyte behaves can be described in terms of the

retention volume = V_r

retention time = t_r

partition ratio = k'

Depending on the mobile and stationary phases chosen, an analyte might be totally retained on a column or migrate without any interaction

υ = the average linear velocity of mobile phase

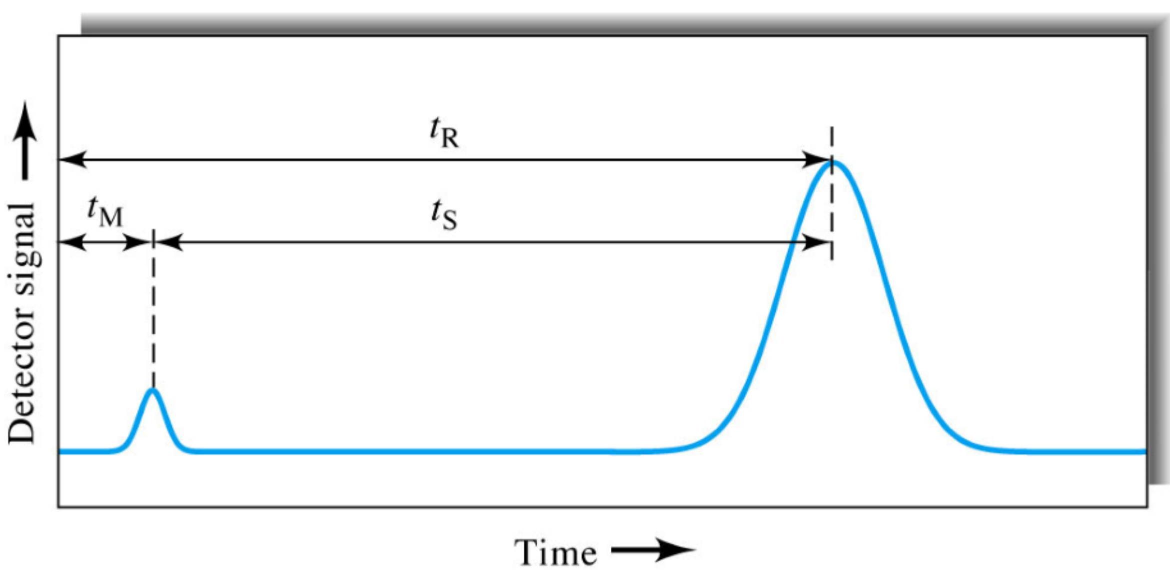
$$\upsilon = \frac{L}{t_m}$$

L = column length (usually cm)

t_m = time for a totally un-retained analyte to pass through the column (dead or void time)

t_s = time analyte spends in the mobile phase

$$t_r = t_s + t_m$$



K= partition coefficient, it characterizes the speed at which an analyte travels down a column.

$$K = \frac{C_s}{C_m}$$

concentrations in the stationary and mobile phases

when K=1 equal distribution between both phases

k' = partition ratio – relates the equilibrium distribution of the sample within the column to the thermodynamic properties of the column (and temperature). k' is a measure of the time spent in the stationary phase compared to the time spent in the mobile phase.

$$A_{mobile} \Leftrightarrow A_{stationary}$$

$$k' = \frac{C_s V_s}{C_m V_m}$$

The partition ratio represents the additional time it takes a retained analyte to elute compared to an un-retained analyte (k'=0) divided by the total retention time of an un-retained band.

$$k' = \frac{(t_r - t_m)}{t_m}$$

Generally k' smaller than one will not provide adequate separation and values greater than ten take too long to elute.

The efficiency of a chromatographic system is expressed as the effective plate number N_{eff}. It indicates the number of times the analyte partitions on its way through the column.

$$N_{eff} = 16\left(\frac{t_r}{W_b}\right)^2$$

W_b = width of the base of the peak

Plate Height = H is the distance the analyte moves while undergoing one partition

$$H = \frac{L}{N_{eff}}$$

A good way to express column efficiency in units of length without specifying the length of the column.

$$N_{eff} = \frac{L}{H}$$

(N_{eff} goes up when H goes down)

1. A capillary column of length L=15 meters was used to separate two organic compounds by gas chromatography. The carrier gas velocity, υ, was measured to be 500 cm min^{-1}. The retention times for the two compounds were 6.00 min (1.26 min) and 8.12 min (1.06 min), respectively. The values in parentheses are the widths (W_b) at the base of each peak.

$$t_r = t_s + t_m$$

a) Calculate the retention time (t_m) for a non-retained solute.

b) Calculate the partition ratio (k') for both compounds.

c) Calculate the effective plate number (N_{eff}) for each compound.

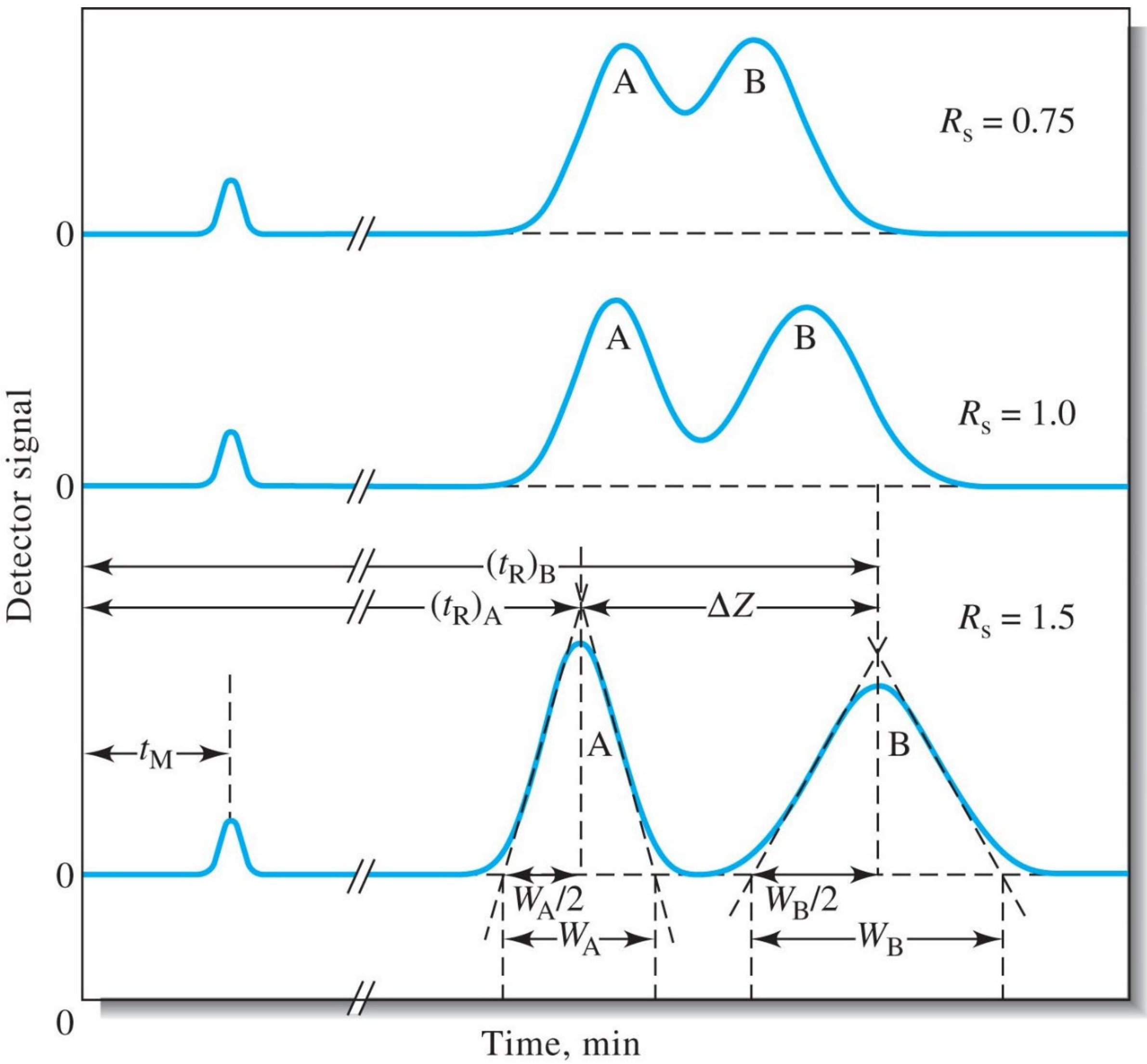
d) Calculate the plate height (H) for each compound in cm.

Column Resolution R_s tells us how far apart two bands are relative to their widths. The resolution of a chromatographic column is a quantitative measure of its ability to separate two analytes A and B.

$$R_s = \frac{2[(t_r)_B - (t_r)_A]}{W_B + W_A}$$

$(t_r)_i$ = retention time

W_i = peak width



2. Substances A and B have retention times of 16.40 and 17.63 minutes, respectively, on a 30.0 cm column. An un-retained species passes through the column in 1.30 min. The peak widths for A and B are 1.11 and 1.21 min, respectively. Calculate the column resolution.

Two general methods for improving the resolution/efficiency of a column

1. **Reduce the plate height (decrease particle size of column packing).**
2. **Increase the column length**

TABLE 31-5

Variables That Influence Column Efficiency

Variable	Symbol	Usual Units
Linear velocity of mobile phase	u	cm s^{-1}
Diffusion coefficient in mobile phase*	D_M	cm^2 s^{-1}
Diffusion coefficient in stationary phase*	D_S	cm^2 s^{-1}
Retention factor (Equation 31-18)	k	unitless
Diameter of packing particles	d_p	cm
Thickness of liquid coating on stationary phase	d_f	cm

*Increases as temperature increases and viscosity decreases.

The variables that lead to band broadening include:

(1) large particle diameters for stationary phases;

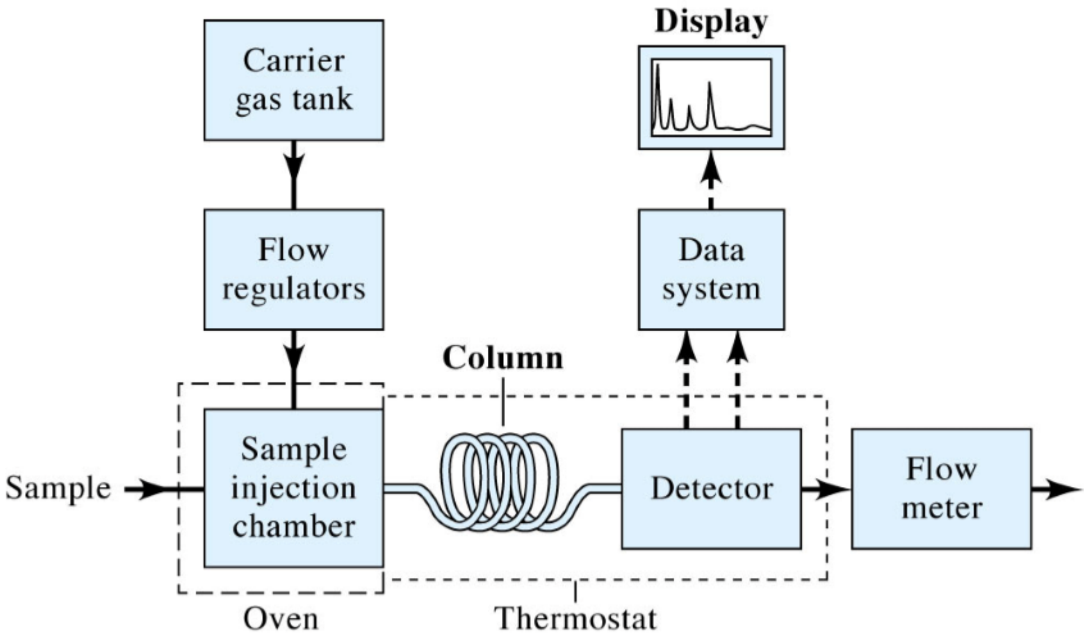
(2) large column diameters;

(3) high temperatures (important only in gas chromatography);

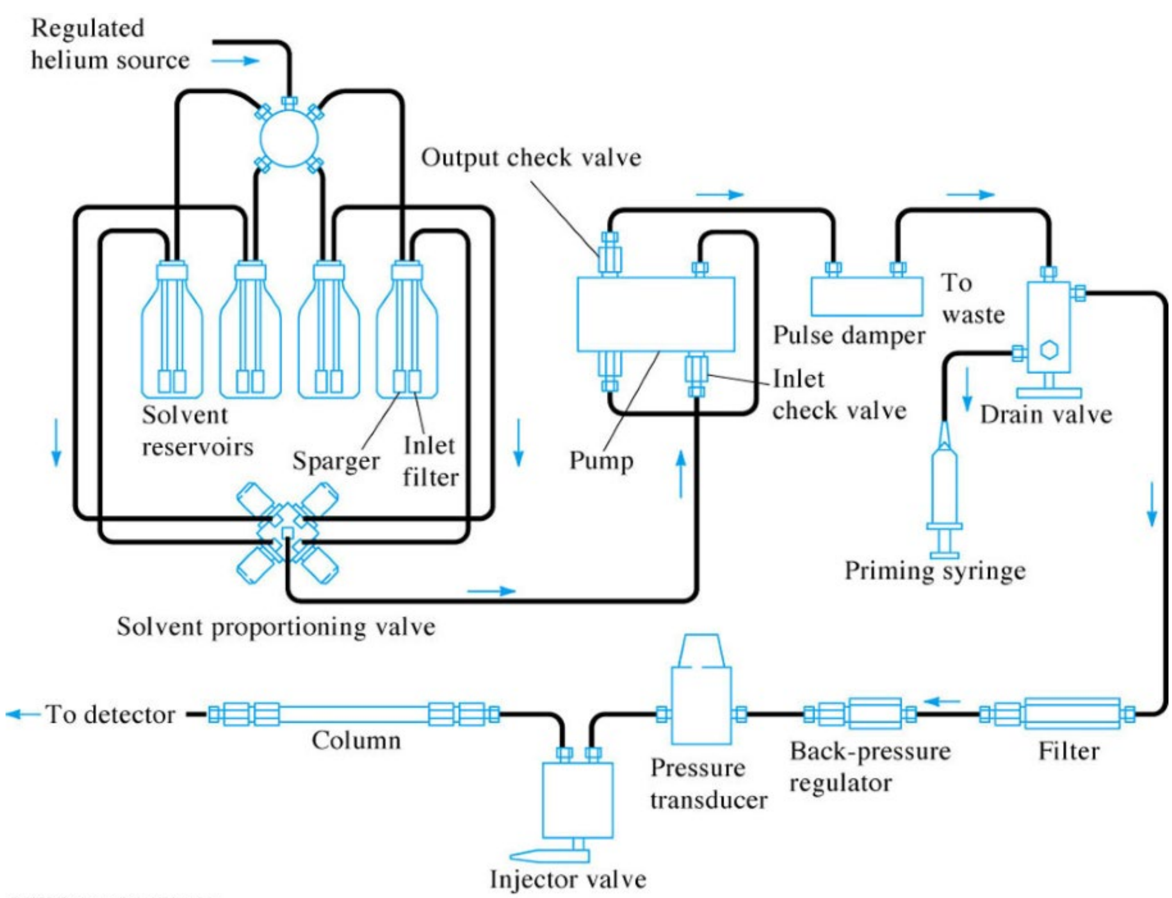
(4) for liquid stationary phases, thick layers of the immobilized liquid;

(5) very rapid or very slow flow rates.

GC


Carrier gas tank
Flow regulators
Display
Data system
Column
Sample
Sample injection chamber
Detector
Flow meter
Oven
Thermostat

HPLC


Regulated helium source
Output check valve
To waste
Pulse damper
Inlet check valve
Drain valve
Solvent reservoirs
Sparger
Inlet filter
Pump
Priming syringe
Solvent proportioning valve
To detector
Column
Pressure transducer
Back-pressure regulator
Filter
Injector valve